

ECON712 Replication Exercise

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Topic: Experimental Estimates of Education Functions

Summary and Contribution

10/15

- **Research question:** The main question of the paper is clearly stated.
- **Contribution:** Your replication exercise discusses the main contribution of the paper to the literature. However, you need to include citations when you claimed that earlier literature does not give a causal answer to the main question because they used observational data, which cannot separate selection from causal effects.
- **Main findings:** Your overview of the main findings is adequate. A point that should have been expanded is how the use of IV identification strategy solves issues with using the project STAR to estimate returns to educations.

Identification Strategy

16/20

- **Identification Problem:** The identification issue is properly discussed, including issues of attenuation bias, omitted variable bias and selection.
- **Identification Strategy:** Your discussion of the empirical strategy is good. In particular, the discussion that refers to the balance tests in Table I as proof that the students were randomly assigned to class size is very clear. On the other hand, you need to either expand on some of the technical concepts you introduced, and were not taught in class, or provide citations. For instance, SUTVA (stable unit treatment value assumption) is a very important concept that I did not teach. Your report does not make clear whether this concept reflects your own understanding or was taken from another source without citation.
- **Empirical Specification:** Your discussion of the empirical specification is adequate. Expanding your discussion across certain areas would have improved the quality of the discussion. For instance, though discussed later in your replication exercise, you do not explain why the author included a school specific fixed-effect term. Also, you do not expand on how obtaining data on initial test scores, before the experiment was performed, would improve the analysis.
- **Critical Assessment:** Your critical assessment is well discussed. I think your discussion of the potential correlation between attrition and potential outcomes is very good. Also, you briefly discuss how the paper uses IV to overcome this issue.

Replication

32/40

- **Data Description:** Your discussion on data sources and data preparation, including discussing potential sources of the discrepancies between your results and the paper's results is excellent. Very good discussion of balance tests and kernel distributions.
- **Main Results:** Overall, the discussion of main results is adequate. There are some areas where you could have provided a more thorough discussion of results. For instance, in the main results table, the difference between OLS and reduced-form estimates should have been further discussed (I introduced the concept of reduced-form estimates within an IV framework in class).

Critical Discussion and Proposed Extensions

23/25

- **Assessment:** Excellent discussion that the class size benefits are solely focused on student test performance, and do not measure other important potential benefits such as labour market outcomes. You should have expanded your discussion on controls improving the efficiency of the estimator, as this was discussion in class (additional controls are correlated with the dependent variable, but uncorrelated with the main regressor due to random assignment).
- **Proposed Improvements:** Good discussion of potential extensions and overall assessment of the paper.

Grade: 81

ECON712 Presentation

- **Quality, organization, and clarity of your presentation (1/1.5) :** The presentation was not well organized. There was also a lot of repetition of certain concepts, while other important topics were not discussed properly (e.g. the IV strategy).
- **Summary of the paper and general discussion of the paper's contribution to the literature (1.5/2):** The main question and importance of the question the paper seeks to answer was well presented. However, there was no thorough discussion of how this paper advances the literature on this topic (for this you needed to provide additional citations).
- **Discussion of the data, methods, results, and identification-related issues (3/4.5):**
 - The discussion of the data, including the variables used in your replication exercise, was adequate. Presenting some summary statistics would have improved the quality of your presentation.
 - Your discussion of the identification strategy and potential issues was appropriate. You discussed issues with not observing baseline test scores before the experiment, but you did not mention how obtaining data on this would improve the analysis.
 - The discussion of the main results of the paper was incomplete. You did not provide enough information in the presentation to properly understand the tables of results in your slides. For instance, there was no discussion of what the different specifications were or why the estimates changed, or did not change, as more controls were added. Also, you did not properly discuss the differences between OLS and reduced-form estimates.
- **Overall assessment of the paper including suggested improvements (1.5/1.5):** Your discussion of the potential improvements to the analysis was good. Your suggestion of tracking students over time to obtain a multi-dimensional measure of the benefits of small class size was very good.
- **Ability to respond to questions and your overall conclusion, including a brief discussion of whether the paper increased your interest in the topic (0/0.5)**

Grade: 7/10